

Data Mining Assignment 4: Classification Readme

Datasets and Preprocessing

All datasets were split in an 80/20 stratified ratio. Missing values were imputed inside the training pipeline.

Numeric outliers were clipped using the 1.5 IQR rule before modeling.

dataset	rows	features	train_rows	test_rows
cervical	858	28	686	172
fetal_health	2126	21	1700	426
banking	41188	20	32950	8238

5-Fold Cross-Validation Mean Metrics

dataset	model	backend	precision_weighted	recall_weighted	f1_weighted	accuracy
banking	AdaBoost	AdaBoostClassifier	0.9037	0.9111	0.9063	0.9111
banking	Decision Tree	DecisionTreeClassifier	0.9049	0.8801	0.8895	0.8895
banking	Random Forest	RandomForestClassifier	0.9183	0.9069	0.9114	0.9069
banking	XGBoost	XGBClassifier	0.9014	0.9122	0.9026	0.9122
cervical	AdaBoost	AdaBoostClassifier	0.8892	0.9038	0.8955	0.9038
cervical	Decision Tree	DecisionTreeClassifier	0.8872	0.8309	0.8566	0.8309
cervical	Random Forest	RandomForestClassifier	0.8906	0.9227	0.9053	0.9227
cervical	XGBoost	XGBClassifier	0.8813	0.9329	0.9055	0.9329
fetal_health	AdaBoost	AdaBoostClassifier	0.9148	0.9106	0.9112	0.9106
fetal_health	Decision Tree	DecisionTreeClassifier	0.9061	0.8894	0.8944	0.8894
fetal_health	Random Forest	RandomForestClassifier	0.9254	0.9259	0.9252	0.9259
fetal_health	XGBoost	XGBClassifier	0.9258	0.9276	0.9256	0.9276

Held-Out Test Metrics

dataset	model	backend	precision_weighted	recall_weighted	f1_weighted	accuracy
cervical	Decision Tree	DecisionTreeClassifier	0.8915	0.8140	0.8483	0.8140
cervical	Random Forest	RandomForestClassifier	0.9020	0.9302	0.9111	0.9302
cervical	XGBoost	XGBClassifier	0.8758	0.9302	0.9022	0.9302
cervical	AdaBoost	AdaBoostClassifier	0.8748	0.9128	0.8934	0.9128
fetal_health	Decision Tree	DecisionTreeClassifier	0.9143	0.9014	0.9055	0.9014
fetal_health	Random Forest	RandomForestClassifier	0.9355	0.9343	0.9348	0.9343
fetal_health	XGBoost	XGBClassifier	0.9442	0.9460	0.9444	0.9460
fetal_health	AdaBoost	AdaBoostClassifier	0.9073	0.8967	0.9004	0.8967
banking	Decision Tree	DecisionTreeClassifier	0.9034	0.8813	0.8898	0.8813
banking	Random Forest	RandomForestClassifier	0.9197	0.9087	0.9131	0.9087
banking	XGBoost	XGBClassifier	0.9043	0.9145	0.9051	0.9145
banking	AdaBoost	AdaBoostClassifier	0.9037	0.9116	0.9063	0.9116

Figures

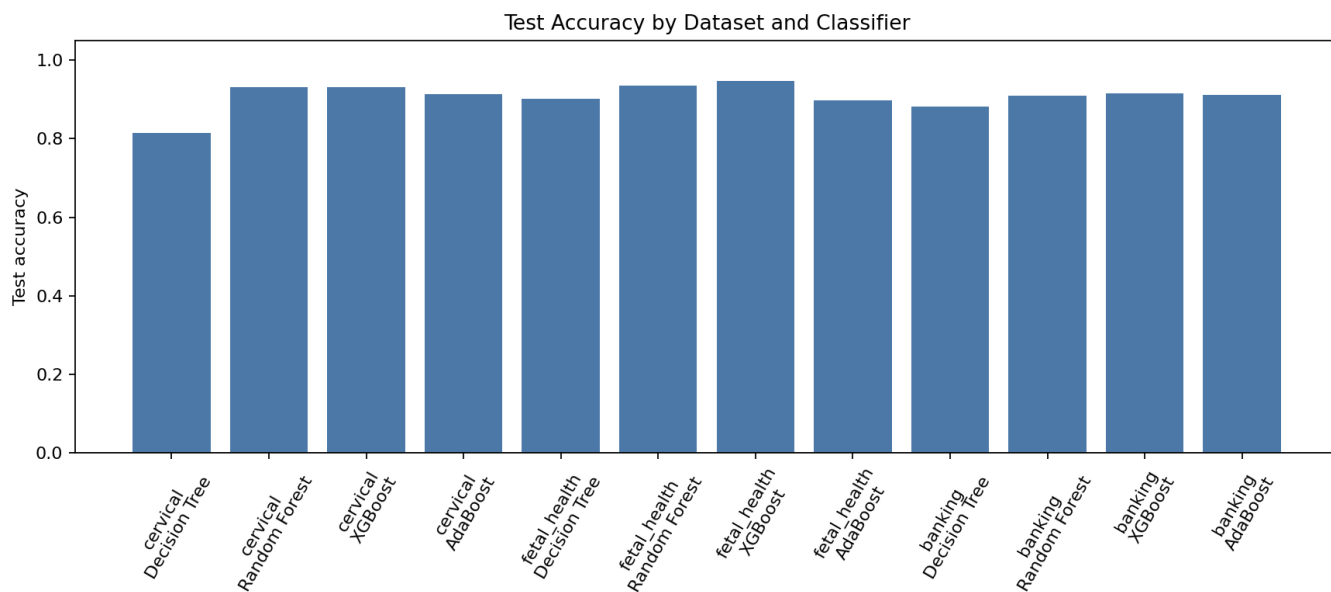


Figure 1: Test accuracy summary.

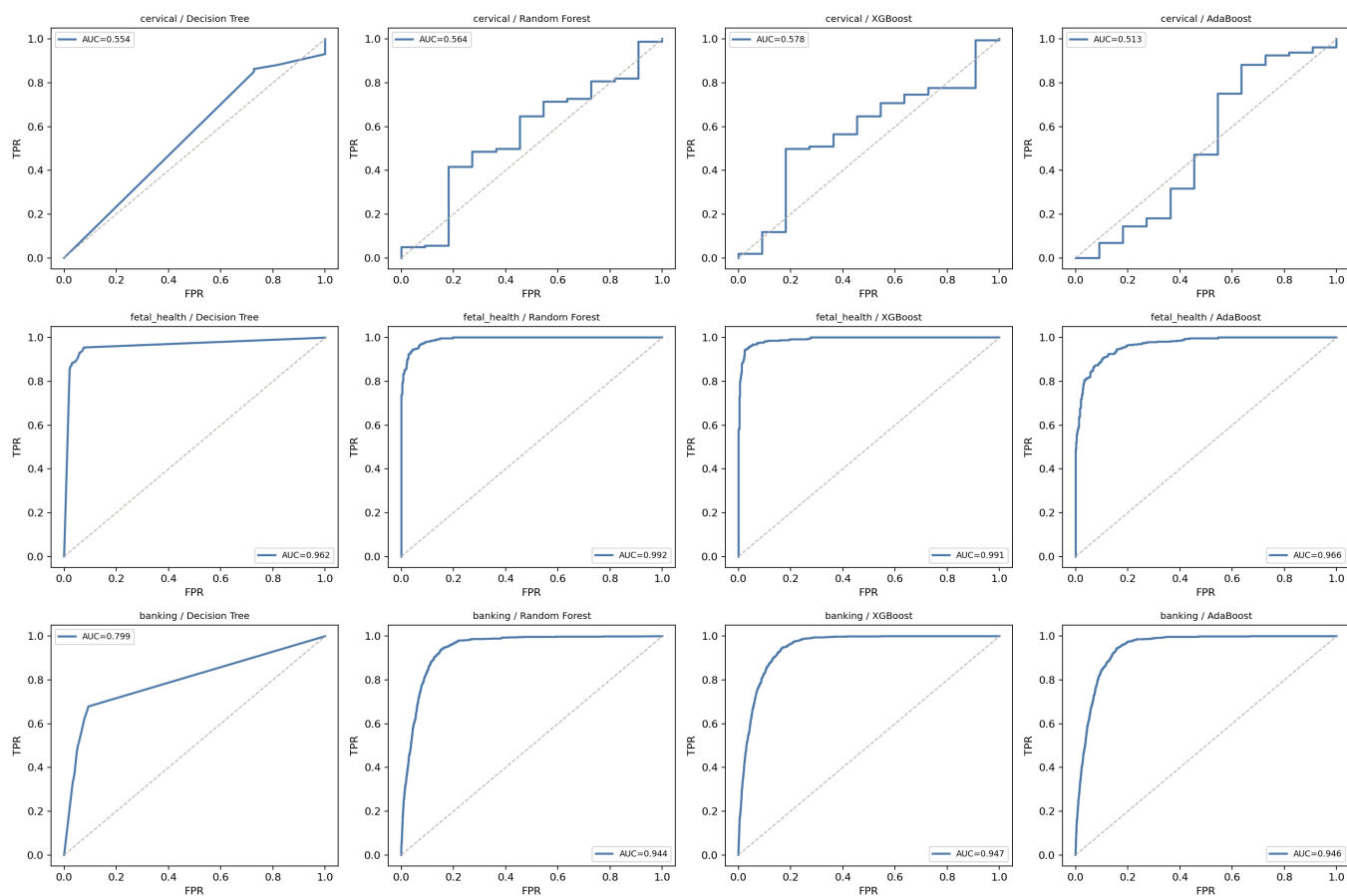


Figure 2: ROC curves.

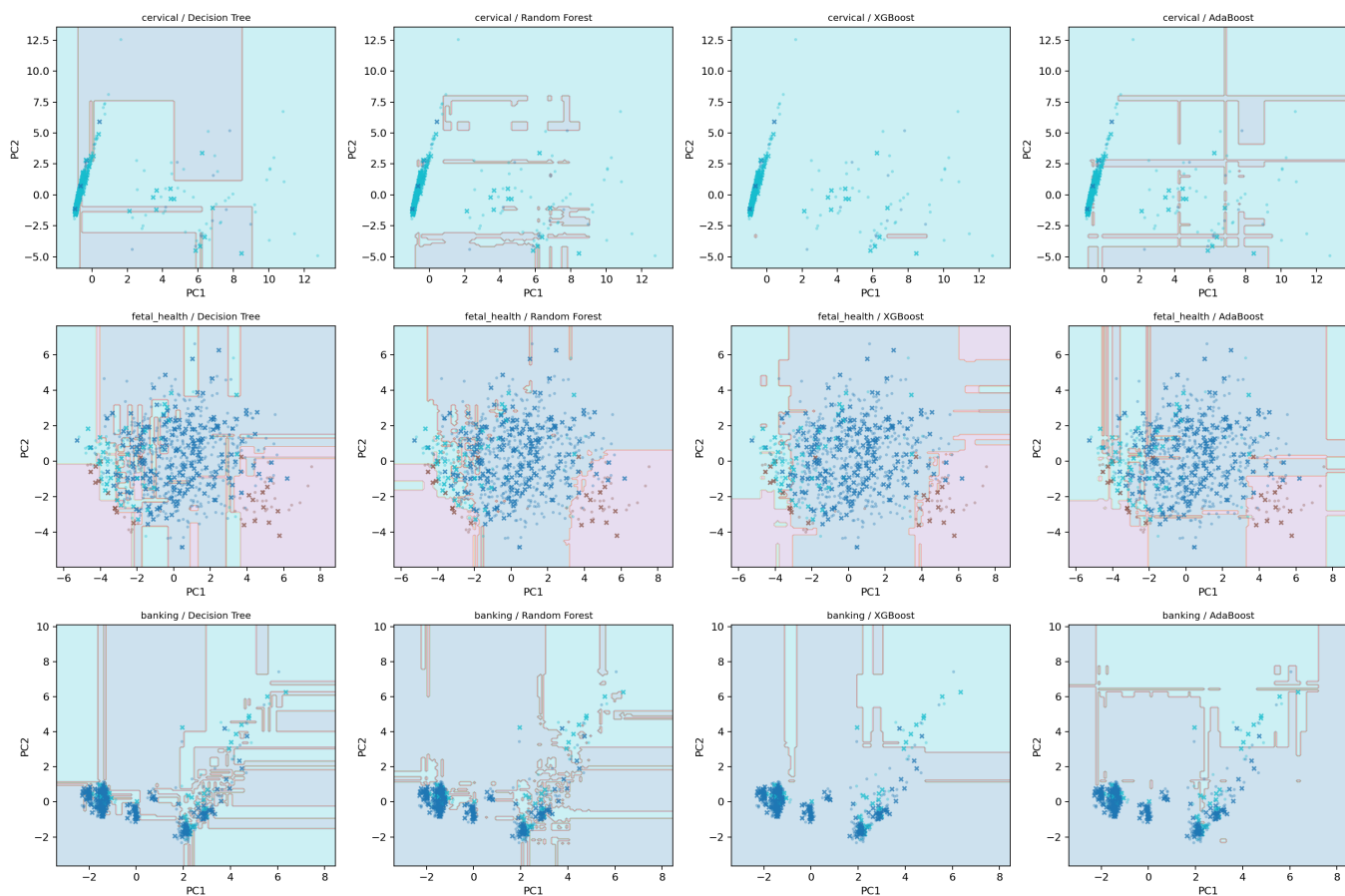


Figure 3: PCA decision-boundary visualizations.